



3N+1 PROBLEM 4

You have successfully solved 3N+1 PROBLEM 4

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The 3N+1 sequence is generated using the following rules: If n is even, the next number is $n / 2$. If n is odd, the next number is $3 \times n + 1$. The sequence continues until it reaches 1. The cycle length of a number is the total number of terms in the sequence, including the starting number and 1.

Given two integers i and j , determine the maximum cycle length among all integers between i and j (inclusive).

If $i > j$, swap them before processing.

Input Format

A single line containing two space-separated integers:

i j

Constraints

$1 \leq i, j \leq 1,000,000$

Output Format

Print three integers:

i j maximum_cycle_length

using the original values of i and j .

Sample Input 0

1 10

Sample Output 0

1 10 20

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Language

Python 3



```
1 import sys
2
3 def solution():
4     input_data = sys.stdin.read().split()
5     if not input_data:
```

```
6         return
7     orig_i = int(input_data[0])
8     orig_j = int(input_data[1])
9
10    low = min(orig_i, orig_j)
11    high = max(orig_i, orig_j)
12
13    memo = {1: 1}
14
15    def get_cycle_length(n):
16        if n in memo:
17            return memo[n]
18
19        orig_n = n
20        count = 0
21        current = n
22        path = []
23
```

Line: 45 Col: 1



Upload Code as File



Test against custom input

Run Code

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Congratulations

You solved this challenge. Would you like to challenge your friends?  

 Test case 0

Compiler Message

 Test case 1 

Success

 Test case 2 